Reinforcement Learning

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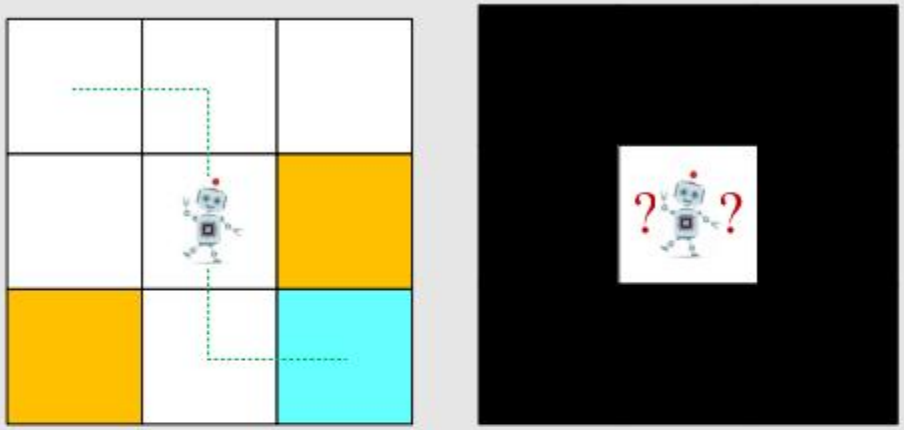
Contents

- First, introduce fundamental concepts in reinforcement learning (RL) by examples.
- Second, formalize the concepts in the context of Markov decision processes.

A grid-world example



An illustrative example used throughout this course:

- Grid of cells: Accessible/forbidden/target cells, boundary.
- Very easy to understand and useful for illustration

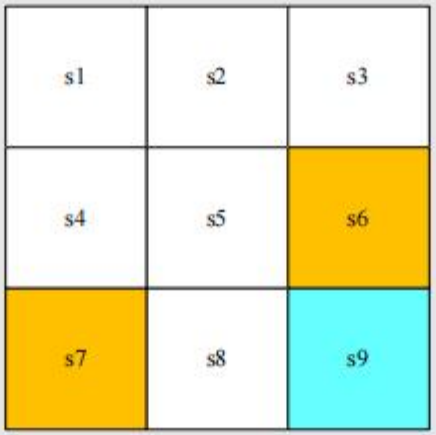
Task:

- Given any starting area, find a "good" way to the target.
- How to define "good"? Avoid forbidden cells, detours, or boundary.

State

State: The status of the agent with respect to the environment.

- For the grid-world example, the location of the agent is the state. There are nine possible locations and hence nine states: $s_1, s_2, \ldots, s_9$.

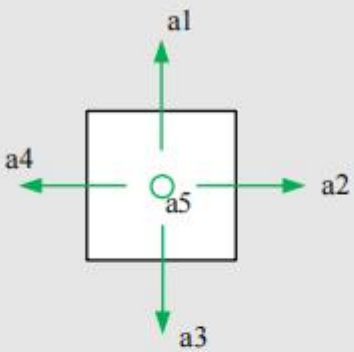


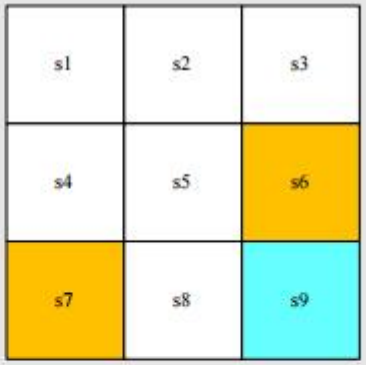
State space: the set of all states $\mathcal{S} = \{s_i\}_{i=1}^{9}$.

Action

Action: For each state, there are five possible actions: $a_1, \ldots, a_5$

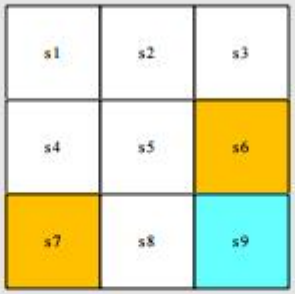
- a_1: move upward;
- a_2: move rightward;
- a_3: move downward;
- a_4: move leftward;
- a_5: stay still;





Action space of a state: the set of all possible actions of a state.
$\mathcal{A}(s_i) = \{a_k\}_{k=1}^{5}$.
Question: can different states have different sets of actions?

State transition



When taking an action, the agent may move from one state to another. Such a process is called *state transition*.

- Example: At state s_1, if we choose action a_2, then what is the next state?

$$s_1 \xrightarrow{a_2} s_2$$

- Example: At state s_1, if we choose action a_1, then what is the next state?

$$s_1 \xrightarrow{a_1} s_1$$

- State transition describes the interaction with the environment.
- Question: Can we define the state transition in other ways? Simulation vs physics

State transition

Pay attention to *forbidden areas*: Example: at state s_5, if we choose action a_2, then what is the next state?

- Case 1: the forbidden area is accessible but with penalty. Then,

$$s_5 \xrightarrow{a_2} s_6$$

- Case 2: the forbidden area is inaccessible (e.g., surrounded by a wall)

$$s_5 \xrightarrow{a_2} s_5$$

We consider the first case, which is more general and challenging.

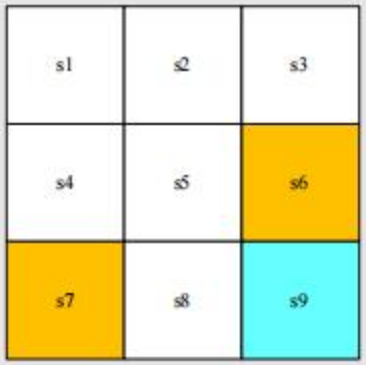
State transition

Tabular representation: We can use a table to describe the state transition:

	a_1 (upward)	a_2 (rightward)	a_3 (downward)	a_4 (leftward)	a_5 (still)
s_1	s_1	s_2	s_4	s_1	s_1
s_2	s_2	s_3	s_5	s_1	s_2
s_3	s_3	s_3	s_6	s_2	s_3
s_4	s_1	s_5	s_7	s_4	s_4
s_5	s_2	s_6	s_8	s_4	s_5
s_6	s_3	s_6	s_9	s_5	s_6
s_7	s_4	s_8	s_7	s_7	s_7
s_8	s_5	s_9	s_8	s_7	s_8
s_9	s_6	s_9	s_9	s_8	s_9

Can only represent *deterministic* cases.

State transition



State transition probability: use probability to describe state transition!

- Intuition: At state s_1, if we choose action a_2, the next state is s_2.
- Math:

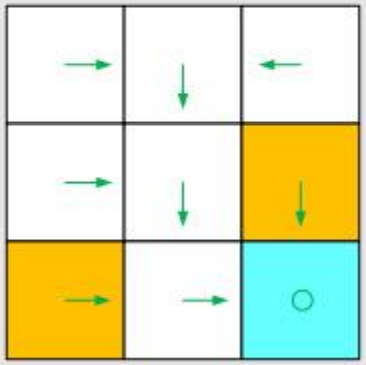
$$p(s_2|s_1, a_2) = 1$$
$$p(s_i|s_1, a_2) = 0 \quad \forall i \neq 2$$

Here it is a deterministic case. The state transition could be stochastic (for example, wind gust).

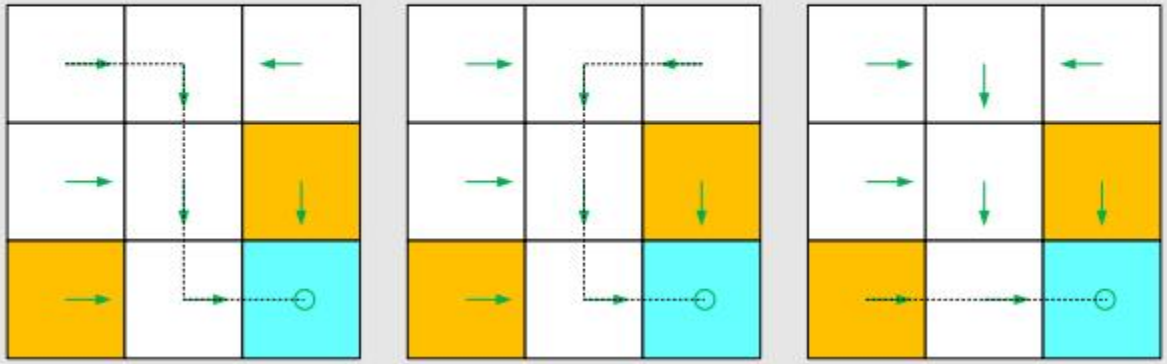
Policy

Policy tells the agent what actions to take at a state.

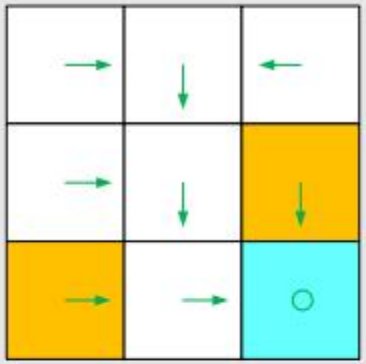
Intuitive representation: We use *arrows* to describe a policy.



Based on this policy, we get the following trajectories with different starting points.



Policy



Mathematical representation: using conditional probability

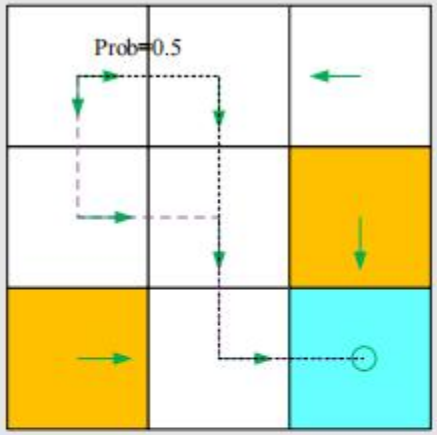
For example, for state s_1:

$$\pi(a_1|s_1) = 0$$
$$\pi(a_2|s_1) = 1$$
$$\pi(a_3|s_1) = 0$$
$$\pi(a_4|s_1) = 0$$
$$\pi(a_5|s_1) = 0$$

It is a deterministic policy.

Policy

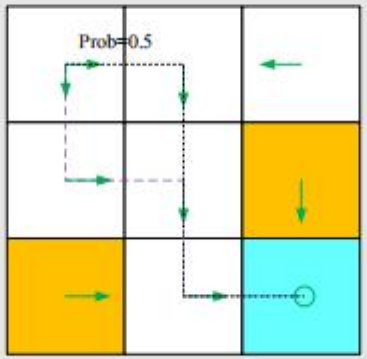
There are stochastic policies.
For example:



In this policy, for s_1:

$$\pi(a_1|s_1) = 0$$
$$\pi(a_2|s_1) = 0.5$$
$$\pi(a_3|s_1) = 0.5$$
$$\pi(a_4|s_1) = 0$$
$$\pi(a_5|s_1) = 0$$

Policy



Tabular representation of a policy: how to use this table.

	a_1 (upward)	a_2 (rightward)	a_3 (downward)	a_4 (leftward)	a_5 (still)
s_1	0	0.5	0.5	0	0
s_2	0	0	1	0	0
s_3	0	0	0	1	0
s_4	0	1	0	0	0
s_5	0	0	1	0	0
s_6	0	0	1	0	0
s_7	0	1	0	0	0
s_8	0	1	0	0	0
s_9	0	0	0	0	1

Can represent either *deterministic* or *stochastic* cases.

Reward

Reward is one of the most unique concepts of RL.

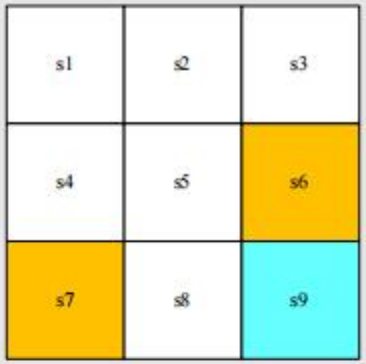
Reward: a real number we get after taking an action.

- A positive reward represents encouragement to take such actions.
- A negative reward represents punishment to take such actions.

Questions:

- Can positive indicate punishment and negative indicate encouragement?
  - Yes.
  - In this case, reward may called *cost*.
- What about a zero reward?
  - Relative values matter, not absolute values.
  - $r = \{+1, -1\}$ becomes $r = \{+2, 0\}$ will not change the optimal policy.

Reward

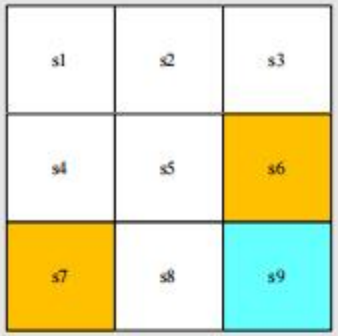


In the grid-world example, the rewards are designed as follows:

- If the agent attempts to get out of the boundary, let $r_{\text{bound}} = -1$
- If the agent attempts to enter a forbidden cell, let $r_{\text{forbid}} = -1$
- If the agent reaches the target cell, let $r_{\text{target}} = +1$
- Otherwise, the agent gets a reward of $r = 0$.

Reward can be interpreted as a **human-machine interface**, with which we can guide the agent to behave as what we expect.
For example, with the above designed rewards, the agent will try to avoid getting out of the boundary or stepping into the forbidden cells.

Reward

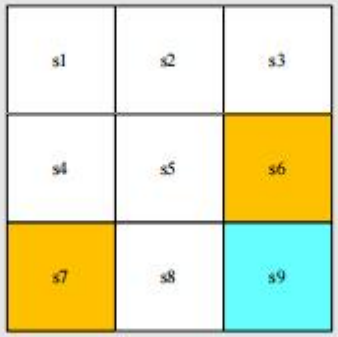


Tabular representation of *reward transition*: how to use the table?

	a_1 (upward)	a_2 (rightward)	a_3 (downward)	a_4 (leftward)	a_5 (still)
s_1	r_{bound}	0	0	r_{bound}	0
s_2	r_{bound}	0	0	0	0
s_3	r_{bound}	r_{bound}	r_{forbid}	0	0
s_4	0	0	r_{forbid}	r_{bound}	0
s_5	0	r_{forbid}	0	0	0
s_6	0	r_{bound}	r_{target}	0	r_{forbid}
s_7	0	0	r_{bound}	r_{bound}	r_{forbid}
s_8	0	r_{target}	r_{bound}	r_{forbid}	0
s_9	r_{forbid}	r_{bound}	r_{bound}	0	r_{target}

Can only represent *deterministic* cases.

Reward



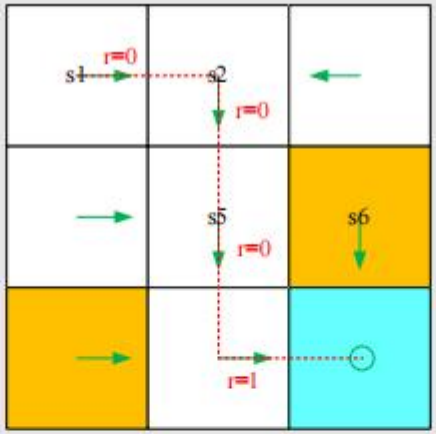
Mathematical description: conditional probability

- Intuition: At state s_1, if we choose action a_1, the reward is -1.
- Math: $p(r = -1|s_1, a_1) = 1$ and $p(r \neq -1|s_1, a_1) = 0$

Remarks:

- Here it is a deterministic case. The reward transition could be stochastic. For example, if you study hard, you will get rewards. But how much is uncertain.

Trajectory and return



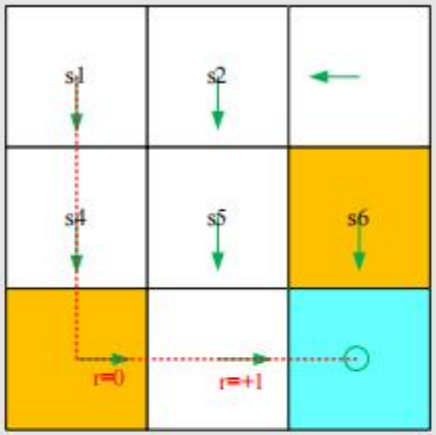
A *trajectory* is a state-action-reward chain:

$$s_1 \xrightarrow[r=0]{a_2} s_2 \xrightarrow[r=0]{a_3} s_5 \xrightarrow[r=0]{a_3} s_8 \xrightarrow[r=1]{a_2} s_9$$

The *return* of this trajectory is the sum of all the rewards collected along the trajectory:

$$\text{return} = 0 + 0 + 0 + 1 = 1$$

Trajectory and return



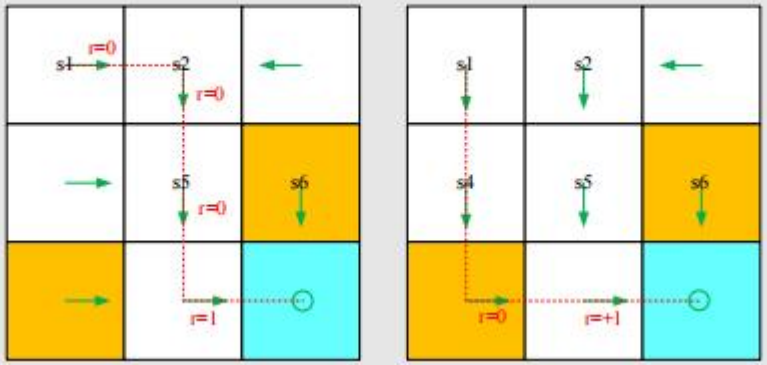
A different policy gives a different trajectory:

$$s_1 \xrightarrow[r=0]{a_3} s_4 \xrightarrow[r=-1]{a_3} s_7 \xrightarrow[r=0]{a_2} s_8 \xrightarrow[r=+1]{a_2} s_9$$

The return of this path is:

$$\text{return} = 0 - 1 + 0 + 1 = 0$$

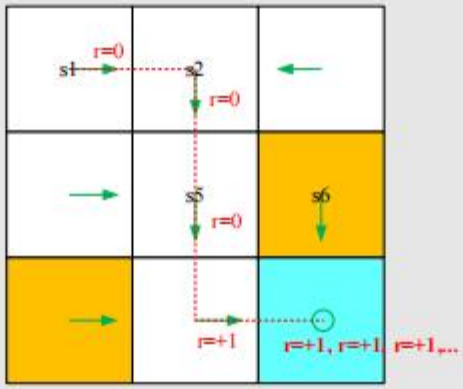
Trajectory and return



Which policy is better?

- **Intuition**: the first is better, because it avoids the forbidden areas.
- **Mathematics**: the first one is better, since it has a greater return!
- Return could be used to evaluate whether a policy is good or not (see details in the next lecture)!

Discounted return



A trajectory may be infinite:

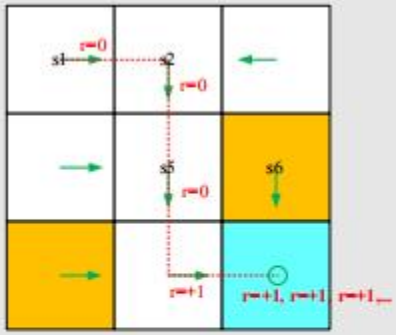
$$s_1 \xrightarrow{a_2} s_2 \xrightarrow{a_3} s_5 \xrightarrow{a_3} s_8 \xrightarrow{a_2} s_9 \xrightarrow{a_5} s_9 \xrightarrow{a_5} s_9 \ldots$$

The return is

$$\text{return} = 0 + 0 + 0 + 1 + 1 + 1 + \ldots = \infty$$

The definition is invalid since the return diverges!

Discounted return



Need to introduce a *discount rate* $\gamma \in (0, 1)$:

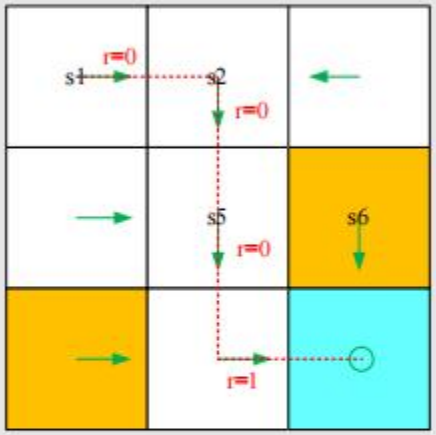
$$\text{discounted return} = 0 + \gamma 0 + \gamma^2 0 + \gamma^3 1 + \gamma^4 1 + \gamma^5 1 + \dots$$
$$= \gamma^3 (1 + \gamma + \gamma^2 + \dots) = \gamma^3 \frac{1}{1-\gamma}.$$

Roles: 1) the sum becomes finite; 2) balance the far and near future rewards:

- If γ is close to 0, the value of the discounted return is dominated by the rewards obtained in the near future.
- If γ is close to 1, the value of the discounted return is dominated by the rewards obtained in the far future.

Episode

When interacting with the environment following a policy, the agent may stop at some *terminal states*. The resulting trajectory is called an *episode* (or a trial).



Example: episode

$$s_1 \xrightarrow[r=0]{a_2} s_2 \xrightarrow[r=0]{a_3} s_5 \xrightarrow[r=0]{a_3} s_8 \xrightarrow[r=1]{a_2} s_9$$

An episode is usually assumed to be a finite trajectory. Tasks with episodes are called *episodic tasks*.

Episode

Some tasks may have no terminal states, meaning the interaction with the environment will never end. Such tasks are called *continuing tasks*.

In the grid-world example, should we stop after arriving the target?

- Treat the target state as a normal state with a policy. The agent can still leave the target state and gain $r = +1$ when entering the target state.
- We don't need to distinguish the target state from the others and can treat it as a normal state.

Markov decision process (MDP)

Key elements of MDP:

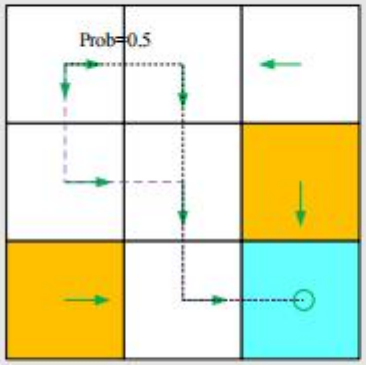
- Sets:
  - State: the set of states $\mathcal{S}$
  - Action: the set of actions $\mathcal{A}(s)$ is associated for state $s \in \mathcal{S}$.
  - Reward: the set of rewards $\mathcal{R}(s, a)$.
- Probability distribution (or called system model):
  - State transition probability: at state s, taking action a, the probability to transit to state s' is $p(s'|s, a)$
  - Reward probability: at state s, taking action a, the probability to get reward r is $p(r|s, a)$
- Policy: at state s, the probability to choose action a is $\pi(a|s)$
- *Markov property*: memoryless property

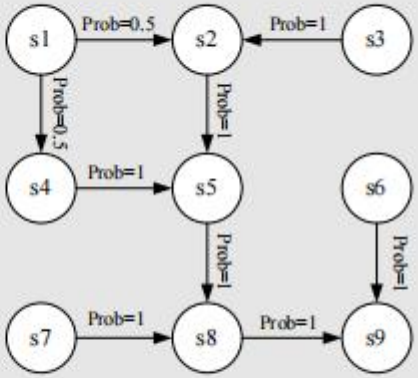
$$p(s_{t+1}|a_t, s_t, \ldots, a_0, s_0) = p(s_{t+1}|a_t, s_t),$$
$$p(r_{t+1}|a_t, s_t, \ldots, a_0, s_0) = p(r_{t+1}|a_t, s_t).$$

All the concepts introduced in this lecture can be put in the framework in MDP.

Markov decision process (MDP)

The grid world could be abstracted as a more general model, *Markov process*.





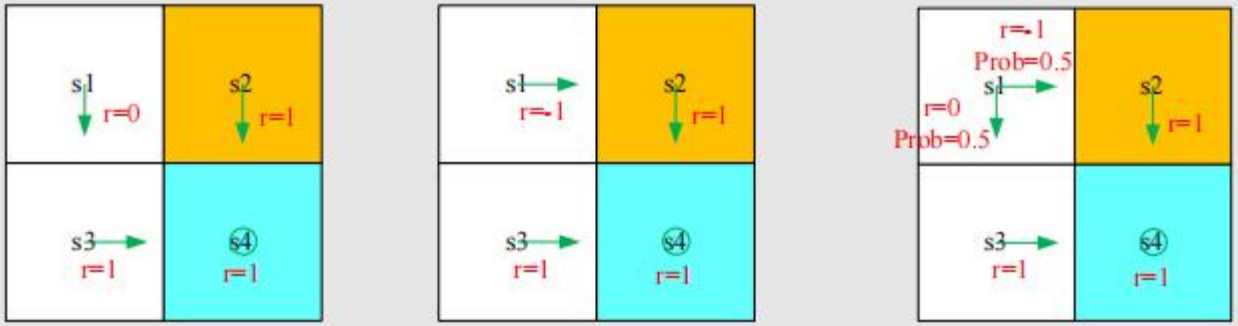
The circles represent states and the links with arrows represent the state transition.

In this lecture:

- A core concept: state value
- A fundamental tool: Bellman equation

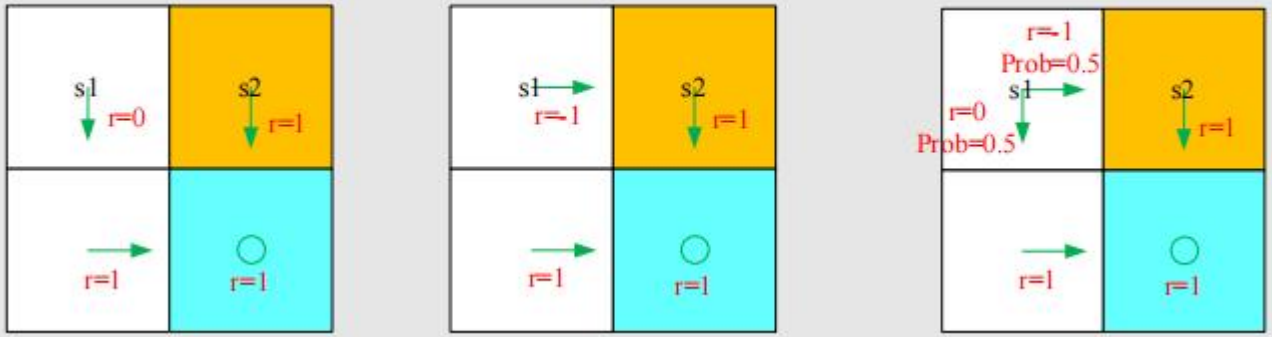
Motivating example 1: Why return is important?

- What is return? The (discounted) sum of the rewards obtained along a trajectory.
- Why is return important? See the following examples.



- Question: From the starting point s_1, which policy is the "best"? Which is the "worst"?
  - Intuition: the first is the best and the second is the worst, because of the forbidden area.
  - Math: can we use mathematics to describe such intuition?
    Return could be used to evaluate policies. See the following.

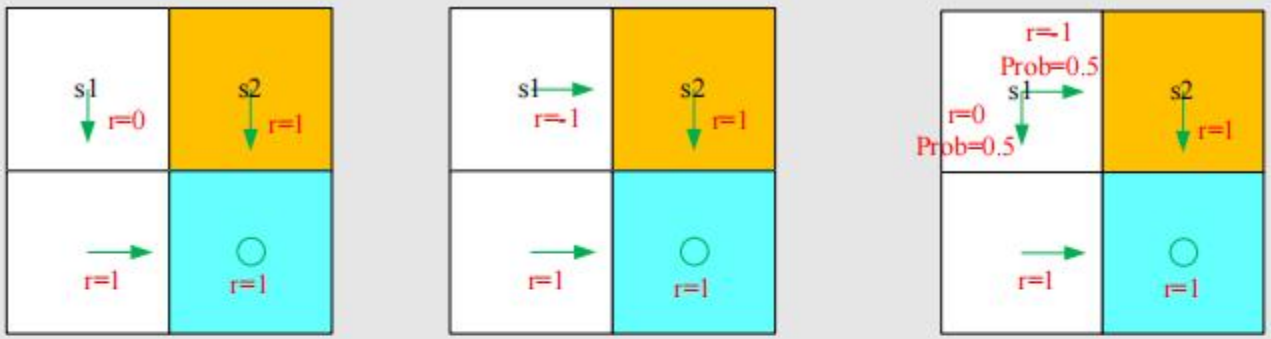
Motivating example 1: Why return is important?



Based on policy 1 (left figure), starting from s_1, the discounted return is

$$\begin{aligned}\text{return}_1 &= 0 + \gamma 1 + \gamma^2 1 + \dots \\ &= \gamma(1 + \gamma + \gamma^2 + \dots) \\ &= \frac{\gamma}{1-\gamma}\end{aligned}$$

Motivating example 1: Why return is important?

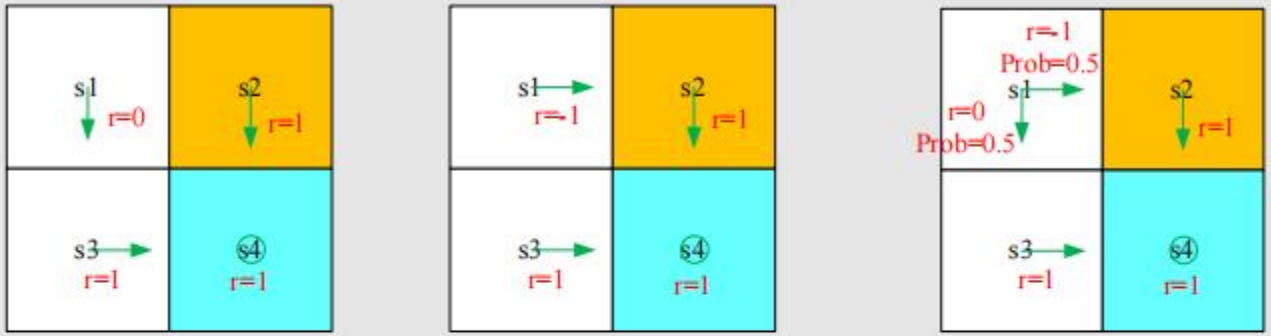


Exercise: Based on policy 2 (middle figure), starting from s_1, what is the discounted return?

Answer:

$$
\begin{aligned}
\text{return}_2 &= -1 + \gamma 1 + \gamma^2 1 + \dots \\
&= -1 + \gamma(1 + \gamma + \gamma^2 + \dots) \\
&= -1 + \frac{\gamma}{1-\gamma}
\end{aligned}
$$

Motivating example 1: Why return is important?

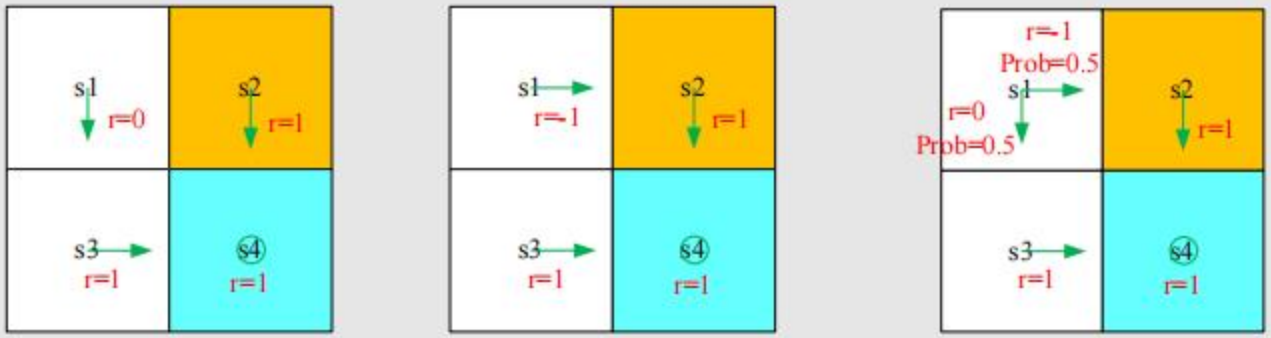


Policy 3 is stochastic!

Exercise: Based on policy 3 (right figure), starting from s_1, the discounted return is

Answer:

$$\begin{aligned}\text{return}_3 &= 0.5\left(-1+\frac{\gamma}{1-\gamma}\right)+0.5\left(\frac{\gamma}{1-\gamma}\right)\\ &= -0.5+\frac{\gamma}{1-\gamma}\end{aligned}$$



In summary, starting from s_1,

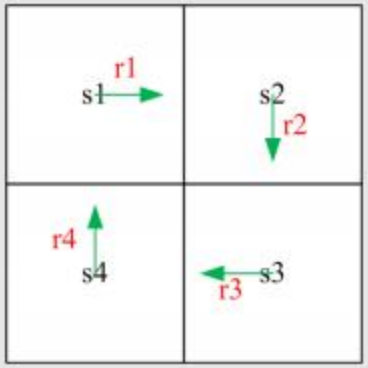
$$\text{return}_1 > \text{return}_3 > \text{return}_2$$

The above inequality suggests that the first policy is the best and the second policy is the worst, which is exactly the same as our intuition.

Calculating return is important to evaluate a policy.

Motivating example 2: How to calculate return?

While return is important, how to calculate it?



Method 1: by definition

Let v_i denote the return obtained starting from s_i $(i = 1, 2, 3, 4)$

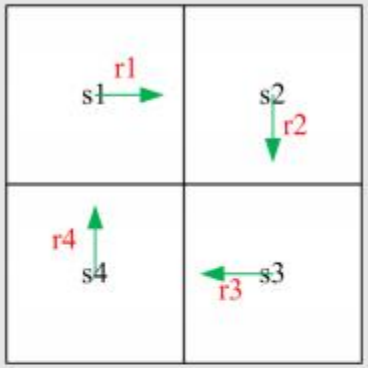
$$v_1 = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots$$

$$v_2 = r_2 + \gamma r_3 + \gamma^2 r_4 + \dots$$

$$v_3 = r_3 + \gamma r_4 + \gamma^2 r_1 + \dots$$

$$v_4 = r_4 + \gamma r_1 + \gamma^2 r_2 + \dots$$

While return is important, how to calculate it?



Method 2:

$$v_1 = r_1 + \gamma(r_2 + \gamma r_3 + \dots) = r_1 + \gamma v_2$$

$$v_2 = r_2 + \gamma(r_3 + \gamma r_4 + \dots) = r_2 + \gamma v_3$$

$$v_3 = r_3 + \gamma(r_4 + \gamma r_1 + \dots) = r_3 + \gamma v_4$$

$$v_4 = r_4 + \gamma(r_1 + \gamma r_2 + \dots) = r_4 + \gamma v_1$$

- The returns rely on each other. *Bootstrapping!*

How to solve these equations? Write in the following matrix-vector form:

$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_{\mathbf{v}} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} + \begin{bmatrix} \gamma v_2 \\ \gamma v_3 \\ \gamma v_4 \\ \gamma v_1 \end{bmatrix} = \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}}_{\mathbf{r}} + \gamma \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{P}} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_{\mathbf{v}}$$

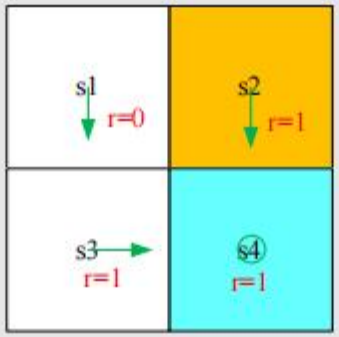
which can be rewritten as

$$\mathbf{v} = \mathbf{r} + \gamma \mathbf{P} \mathbf{v}$$

This is the Bellman equation (for this specific deterministic problem)!!

- Though simple, it demonstrates the core idea: the value of one state relies on the values of other states.
- A matrix-vector form is more clear to see how to solve the state values.

Exercise: Consider the policy shown in the figure. Please write out the relation among the returns (that is to write out the Bellman equation)



Answer:

$$v_1 = 0 + \gamma v_3$$
$$v_2 = 1 + \gamma v_4$$
$$v_3 = 1 + \gamma v_4$$
$$v_4 = 1 + \gamma v_4$$

Exercise: How to solve them? We can first calculate v_4, and then v_3, v_2, v_1.

Consider the following single-step process:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1}$$

- $t, t+1$: discrete time instances
- S_t: state at time t
- A_t: the action taken in state S_t
- R_{t+1}: the reward obtained after taking A_t
- S_{t+1}: the state transited to after taking A_t

Note that S_t, A_t, R_{t+1} are all *random variables*.
This step is governed by the following probability distributions:

- $S_t \to A_t$ is governed by $\pi(A_t = a | S_t = s)$
- $S_t, A_t \to R_{t+1}$ is governed by $p(R_{t+1} = r | S_t = s, A_t = a)$
- $S_t, A_t \to S_{t+1}$ is governed by $p(S_{t+1} = s' | S_t = s, A_t = a)$

At this moment, we assume we know the model (i.e., the probability distributions)!

Some notations

Consider the following multi-step trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \ldots$$

The discounted return is

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots$$

- $\gamma \in (0, 1)$ is a discount rate.
- G_t is also a random variable since $R_{t+1}, R_{t+2}, \ldots$ are random variables.

State value

The expectation (or called expected value or mean) of G_t is defined as the *state-value function* or simply *state value*:

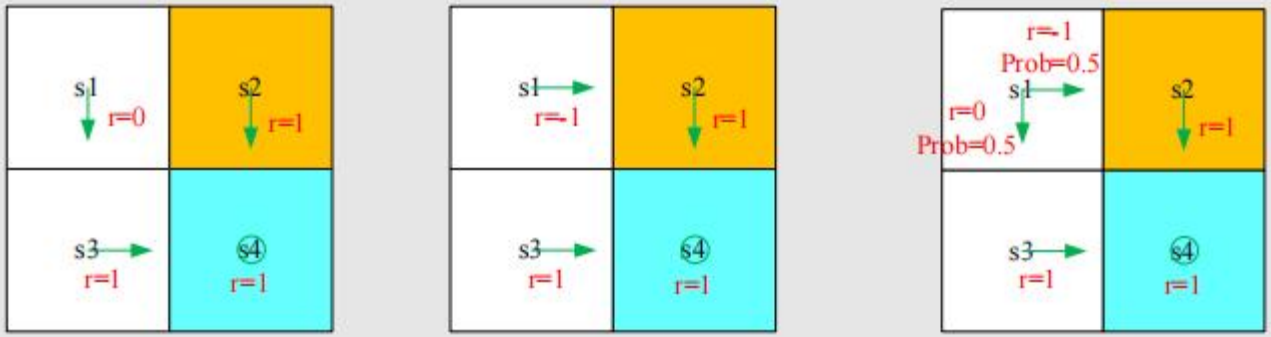
$$v_\pi(s) = \mathbb{E}[G_t | S_t = s]$$

Remarks:

- It is a function of s. It is a conditional expectation with the condition that the state starts from s.
- It is based on the policy π. For a different policy, the state value may be different.

Q: What is the relationship between return and state value?
A: The state value is the mean of all possible returns that can be obtained starting from a state. If everything - $\pi(a|s)$, $p(r|s,a)$, $p(s'|s,a)$ - is deterministic, then state value is the same as return.

Example: which policy is good, which is bad?



Recall the returns obtained from s_1 for the three examples:

$$v_{\pi_1}(s_1) = 0 + \gamma 1 + \gamma^2 1 + \cdots = \gamma(1 + \gamma + \gamma^2 + \ldots) = \frac{\gamma}{1-\gamma}$$

$$v_{\pi_2}(s_1) = -1 + \gamma 1 + \gamma^2 1 + \cdots = -1 + \gamma(1 + \gamma + \gamma^2 + \ldots) = -1 + \frac{\gamma}{1-\gamma}$$

$$v_{\pi_3}(s_1) = 0.5\left(-1 + \frac{\gamma}{1-\gamma}\right) + 0.5\left(\frac{\gamma}{1-\gamma}\right) = -0.5 + \frac{\gamma}{1-\gamma}$$

Bellman equation

- While state value is important, how to calculate? The answer lies in the Bellman equation.
- In a word, the Bellman equation describes the relationship among the values of all states.
- Next, we derive the Bellman equation.
  - There is some math.
  - We already have the intuition.

Deriving the Bellman equation

Consider a random trajectory:

$$S_t \xrightarrow{A_t} R_{t+1}, S_{t+1} \xrightarrow{A_{t+1}} R_{t+2}, S_{t+2} \xrightarrow{A_{t+2}} R_{t+3}, \ldots$$

The return G_t can be written as

$$\begin{aligned} G_t &= R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots, \\ &= R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \ldots), \\ &= R_{t+1} + \gamma G_{t+1}, \end{aligned}$$

Then, it follows from the definition of the state value that

$$\begin{aligned} v_\pi(s) &= \mathbb{E}[G_t | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \mathbb{E}[R_{t+1} | S_t = s] + \gamma \mathbb{E}[G_{t+1} | S_t = s] \end{aligned}$$

Next, calculate the two terms, respectively.

First, calculate the first term $\mathbb{E}[R_{t+1}|S_t = s]$:

$$\begin{aligned}\mathbb{E}[R_{t+1}|S_t = s] &= \sum_a \pi(a|s)\mathbb{E}[R_{t+1}|S_t = s, A_t = a] \\ &= \sum_a \pi(a|s) \sum_r p(r|s,a)r\end{aligned}$$

Note that

- This is the mean of *immediate rewards*

Second, calculate the second term $\mathbb{E}[G_{t+1}|S_t = s]$:

$$\begin{aligned}
\mathbb{E}[G_{t+1}|S_t = s] &= \sum_{s'} \mathbb{E}[G_{t+1}|S_t = s, S_{t+1} = s']p(s'|s) \\
&= \sum_{s'} \mathbb{E}[G_{t+1}|S_{t+1} = s']p(s'|s) \\
&= \sum_{s'} v_\pi(s')p(s'|s) \\
&= \sum_{s'} v_\pi(s') \sum_a p(s'|s,a)\pi(a|s)
\end{aligned}$$

Note that

- This is the mean of *future rewards*
- $\mathbb{E}[G_{t+1}|S_t = s, S_{t+1} = s'] = \mathbb{E}[G_{t+1}|S_{t+1} = s']$ due to the memoryless Markov property.

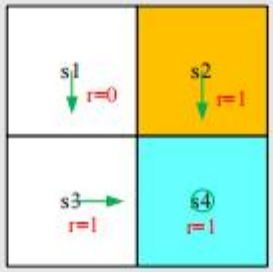
Therefore, we have

$$
\begin{aligned}
v_\pi(s) &= \mathbb{E}[R_{t+1}|S_t = s] + \gamma \mathbb{E}[G_{t+1}|S_t = s], \\
&= \underbrace{\sum_a \pi(a|s) \sum_r p(r|s,a) r}_{\text{mean of immediate rewards}} + \underbrace{\gamma \sum_a \pi(a|s) \sum_{s'} p(s'|s,a) v_\pi(s')}_{\text{mean of future rewards}}, \\
&= \sum_a \pi(a|s) \left[\sum_r p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \right], \quad \forall s \in \mathcal{S}.
\end{aligned}
$$

Highlights:

- The above equation is called the *Bellman equation*, which characterizes the relationship among the state-value functions of different states.
- It consists of two terms: the immediate reward term and the future reward term.
- A set of equations: every state has an equation like this!!!

An illustrative example



Write out the Bellman equation according to the general expression:

$$v_\pi(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \right]$$

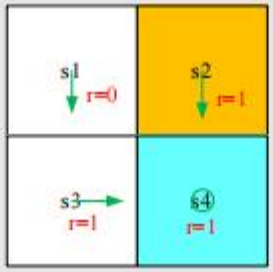
This example is simple because the policy is deterministic.

First, consider the state value of s_1:

- $\pi(a = a_3|s_1) = 1$ and $\pi(a \neq a_3|s_1) = 0$.
- $p(s' = s_3|s_1, a_3) = 1$ and $p(s' \neq s_3|s_1, a_3) = 0$.
- $p(r = 0|s_1, a_3) = 1$ and $p(r \neq 0|s_1, a_3) = 0$.

Substituting them into the Bellman equation gives

$$v_\pi(s_1) = 0 + \gamma v_\pi(s_3)$$



Write out the Bellman equation according to the general expression:

$$v_\pi(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \right]$$

Similarly, it can be obtained that

$$\begin{aligned}
v_\pi(s_1) &= 0 + \gamma v_\pi(s_3), \\
v_\pi(s_2) &= 1 + \gamma v_\pi(s_4), \\
v_\pi(s_3) &= 1 + \gamma v_\pi(s_4), \\
v_\pi(s_4) &= 1 + \gamma v_\pi(s_4).
\end{aligned}$$

How to solve them?

$$
\begin{aligned}
v_\pi(s_1) &= 0 + \gamma v_\pi(s_3), \\
v_\pi(s_2) &= 1 + \gamma v_\pi(s_4), \\
v_\pi(s_3) &= 1 + \gamma v_\pi(s_4), \\
v_\pi(s_4) &= 1 + \gamma v_\pi(s_4).
\end{aligned}
$$

Solve the above equations one by one from the last to the first:

$$
\begin{aligned}
v_\pi(s_4) &= \frac{1}{1-\gamma}, \\
v_\pi(s_3) &= \frac{1}{1-\gamma}, \\
v_\pi(s_2) &= \frac{1}{1-\gamma}, \\
v_\pi(s_1) &= \frac{\gamma}{1-\gamma}.
\end{aligned}
$$

If $\gamma = 0.9$, then

$$v_\pi(s_4) = \frac{1}{1-0.9} = 10,$$
$$v_\pi(s_3) = \frac{1}{1-0.9} = 10,$$
$$v_\pi(s_2) = \frac{1}{1-0.9} = 10,$$
$$v_\pi(s_1) = \frac{0.9}{1-0.9} = 9.$$

What to do after we have calculated state values? Be patient (calculating action value and improve policy)

Matrix-vector form of the Bellman equation

Why consider the matrix-vector form? Because we need to solve the state values from it!

- One unknown relies on another unknown. How to solve the unknowns?

$$v_\pi(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \right]$$

- Elementwise form: The above *elementwise form* is valid for every state $s \in \mathcal{S}$. That means there are $|\mathcal{S}|$ equations like this!
- Matrix-vector form: If we put all the equations together, we have a set of linear equations, which can be concisely written in a *matrix-vector form*. The matrix-vector form is very elegant and important.

Recall that:

$$v_\pi(s) = \sum_a \pi(a|s) \left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \right]$$

Rewrite the Bellman equation as

$$v_\pi(s) = r_\pi(s) + \gamma \sum_{s'} p_\pi(s'|s) v_\pi(s') \tag{1}$$

where

$$r_\pi(s) \triangleq \sum_a \pi(a|s) \sum_r p(r|s,a)r, \qquad p_\pi(s'|s) \triangleq \sum_a \pi(a|s) p(s'|s,a)$$

Suppose the states could be indexed as s_i $(i = 1, \ldots, n)$.
For state s_i, the Bellman equation is

$$v_\pi(s_i) = r_\pi(s_i) + \gamma \sum_{s_j} p_\pi(s_j|s_i) v_\pi(s_j)$$

Put all these equations for all the states together and rewrite to a matrix-vector form

$$v_\pi = r_\pi + \gamma P_\pi v_\pi$$

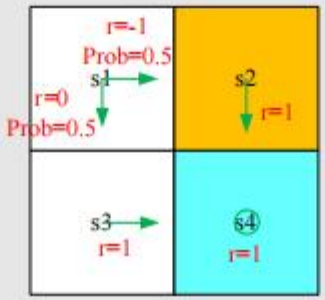
where

- $v_\pi = [v_\pi(s_1), \ldots, v_\pi(s_n)]^T \in \mathbb{R}^n$
- $r_\pi = [r_\pi(s_1), \ldots, r_\pi(s_n)]^T \in \mathbb{R}^n$
- $P_\pi \in \mathbb{R}^{n \times n}$, where $[P_\pi]_{ij} = p_\pi(s_j|s_i)$, is the *state transition matrix*

Illustrative examples

If there are four states, $v_\pi = r_\pi + \gamma P_\pi v_\pi$ can be written out as

$$\underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi} = \underbrace{\begin{bmatrix} r_\pi(s_1) \\ r_\pi(s_2) \\ r_\pi(s_3) \\ r_\pi(s_4) \end{bmatrix}}_{r_\pi} + \gamma \underbrace{\begin{bmatrix} p_\pi(s_1|s_1) & p_\pi(s_2|s_1) & p_\pi(s_3|s_1) & p_\pi(s_4|s_1) \\ p_\pi(s_1|s_2) & p_\pi(s_2|s_2) & p_\pi(s_3|s_2) & p_\pi(s_4|s_2) \\ p_\pi(s_1|s_3) & p_\pi(s_2|s_3) & p_\pi(s_3|s_3) & p_\pi(s_4|s_3) \\ p_\pi(s_1|s_4) & p_\pi(s_2|s_4) & p_\pi(s_3|s_4) & p_\pi(s_4|s_4) \end{bmatrix}}_{P_\pi} \underbrace{\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}}_{v_\pi}.$$



For this specific example:

$$\begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix} = \begin{bmatrix} 0.5(0) + 0.5(-1) \\ 1 \\ 1 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_\pi(s_1) \\ v_\pi(s_2) \\ v_\pi(s_3) \\ v_\pi(s_4) \end{bmatrix}.$$

Solve state value

Why to solve state values?

- Given a policy, finding out the corresponding state values is called *policy evaluation*!
- It is a fundamental problem in RL. It is the foundation to find better policies.
- Therefore, it is important to understand how to solve the Bellman equation.

The Bellman equation in matrix-vector form is

$$v_\pi = r_\pi + \gamma P_\pi v_\pi$$

- The *closed-form solution* is:

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi$$

  - The matrix $I - \gamma P_\pi$ is inevitable. See details in my book.
  - We still need to use numerical algorithms to calculate the matrix inverse.
  - Can we avoid the matrix inverse operation? Yes, as shown below.

- *An iterative solution is:*

$$v_{k+1} = r_\pi + \gamma P_\pi v_k$$

This algorithm leads to a sequence $\{v_0, v_1, v_2, \dots\}$. We can show that

$$v_k \to v_\pi = (I - \gamma P_\pi)^{-1} r_\pi, \quad k \to \infty$$

Proof.

Define the error as $\delta_k = v_k - v_\pi$. We only need to show $\delta_k \to 0$. Substituting $v_{k+1} = \delta_{k+1} + v_\pi$ and $v_k = \delta_k + v_\pi$ into $v_{k+1} = r_\pi + \gamma P_\pi v_k$ gives

$$\delta_{k+1} + v_\pi = r_\pi + \gamma P_\pi(\delta_k + v_\pi),$$

which can be rewritten as

$$\delta_{k+1} = -v_\pi + r_\pi + \gamma P_\pi \delta_k + \gamma P_\pi v_\pi = \gamma P_\pi \delta_k.$$

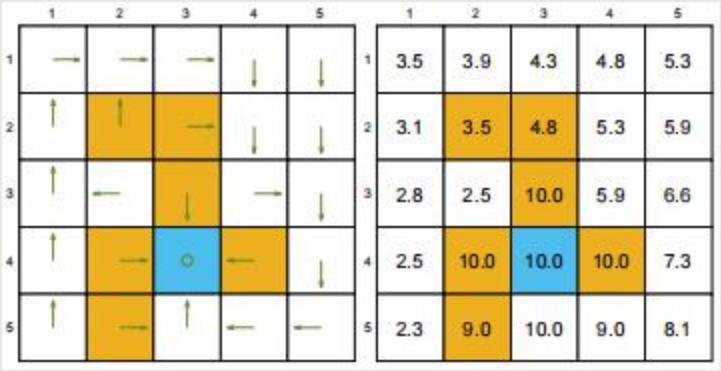
As a result,

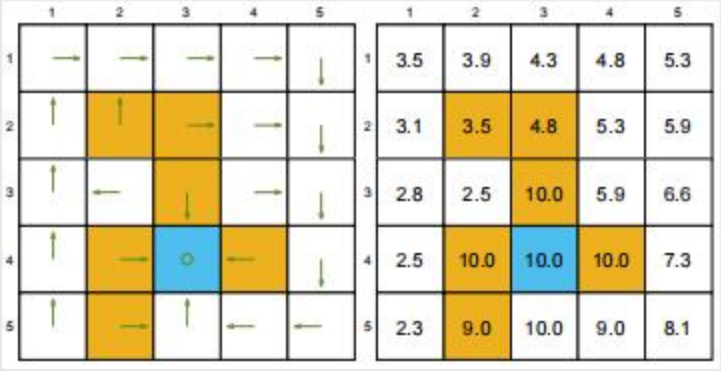
$$\delta_{k+1} = \gamma P_\pi \delta_k = \gamma^2 P_\pi^2 \delta_{k-1} = \cdots = \gamma^{k+1} P_\pi^{k+1} \delta_0.$$

Note that $0 \leq P_\pi^k \leq 1$, which means every entry of P_π^k is no greater than 1 for any $k = 0, 1, 2, \ldots$. That is because $P_\pi^k \mathbf{1} = \mathbf{1}$, where $\mathbf{1} = [1, \ldots, 1]^T$. On the other hand, since $\gamma < 1$, we know $\gamma^k \to 0$ and hence $\delta_{k+1} = \gamma^{k+1} P_\pi^{k+1} \delta_0 \to 0$ as $k \to \infty$. $\square$

Examples: $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, $r_{\text{target}} = +1$, $\gamma = 0.9$

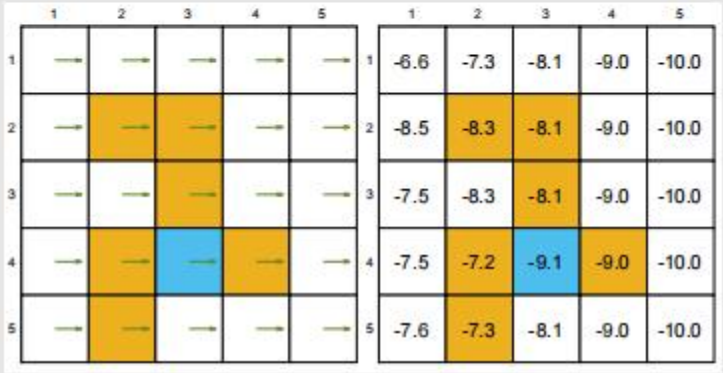
- The following are two "good" policies and the state values. The two policies are different for the top two states in the forth column.



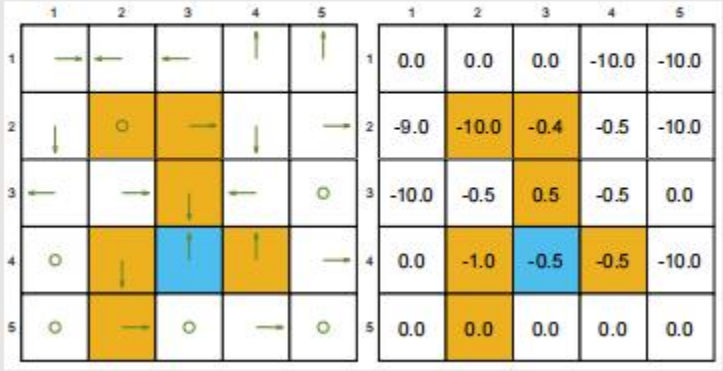


Examples: $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, $r_{\text{target}} = +1$, $\gamma = 0.9$

- The following are two "bad" policies and the state values. The state values are less than those of the good policies.



	1	2	3	4	5
1	-6.6	-7.3	-8.1	-9.0	-10.0
2	-8.5	-8.3	-8.1	-9.0	-10.0
3	-7.5	-8.3	-8.1	-9.0	-10.0
4	-7.5	-7.2	-9.1	-9.0	-10.0
5	-7.6	-7.3	-8.1	-9.0	-10.0



	1	2	3	4	5
1	0.0	0.0	0.0	-10.0	-10.0
2	-9.0	-10.0	-0.4	-0.5	-10.0
3	-10.0	-0.5	0.5	-0.5	0.0
4	0.0	-1.0	-0.5	-0.5	-10.0
5	0.0	0.0	0.0	0.0	0.0

Action value

From state value to action value:

- State value: the average return the agent can get *starting from a state*.
- Action value: the average return the agent can get *starting from a state* and *taking an action*.

Why do we care action value? Because we want to know which action is better. This point will be clearer in the following lectures.

We will frequently use action values.

Definition:

$$q_\pi(s,a) = \mathbb{E}[G_t | S_t = s, A_t = a]$$

- $q_\pi(s,a)$ is a function of the state-action pair (s,a)
- $q_\pi(s,a)$ depends on π

It follows from the properties of conditional expectation that

$$\underbrace{\mathbb{E}[G_t | S_t = s]}_{v_\pi(s)} = \sum_a \underbrace{\mathbb{E}[G_t | S_t = s, A_t = a]}_{q_\pi(s,a)} \pi(a|s)$$

Hence,

$$v_\pi(s) = \sum_a \pi(a|s) q_\pi(s,a) \tag{2}$$

Recall that the state value is given by

$$v_\pi(s) = \sum_a \pi(a|s) \underbrace{\left[\sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \right]}_{q_\pi(s,a)} \tag{3}$$

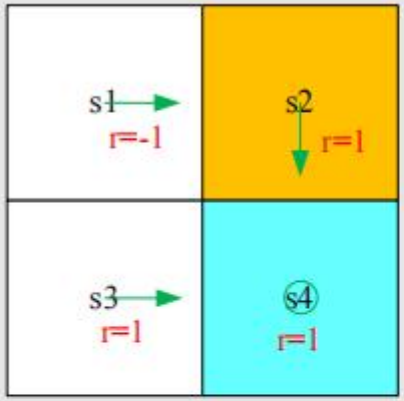
By comparing (2) and (3), we have the **action-value function** as

$$q_\pi(s,a) = \sum_r p(r|s,a)r + \gamma \sum_{s'} p(s'|s,a) v_\pi(s') \tag{4}$$

(2) and (4) are the two sides of the same coin:

- (2) shows how to obtain state values from action values.
- (4) shows how to obtain action values from state values.

Illustrative example for action value

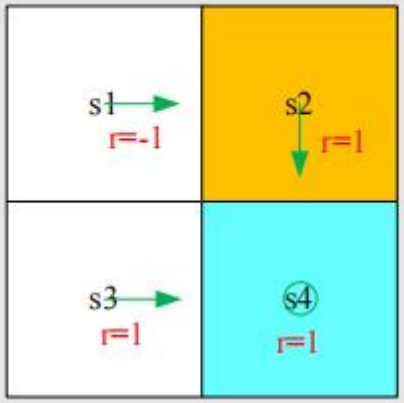


Write out the action values for state s_1.

$$q_\pi(s_1, a_2) = -1 + \gamma v_\pi(s_2),$$

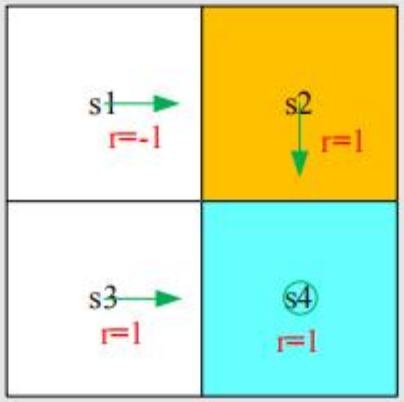
Questions:

- $q_\pi(s_1, a_1), q_\pi(s_1, a_3), q_\pi(s_1, a_4), q_\pi(s_1, a_5) =?$ Be careful!



For the other actions:

$$q_\pi(s_1, a_1) = -1 + \gamma v_\pi(s_1),$$
$$q_\pi(s_1, a_3) = 0 + \gamma v_\pi(s_3),$$
$$q_\pi(s_1, a_4) = -1 + \gamma v_\pi(s_1),$$
$$q_\pi(s_1, a_5) = 0 + \gamma v_\pi(s_1).$$



Highlights:

- Action value is important since we care about which action to take.
- We can first calculate all the state values and then calculate the action values.
- We can also directly calculate the action values with or without models.